Exp. No: 16

Fiber-Optic Link Design, Link Budget and BER Analysis using OptiSystem

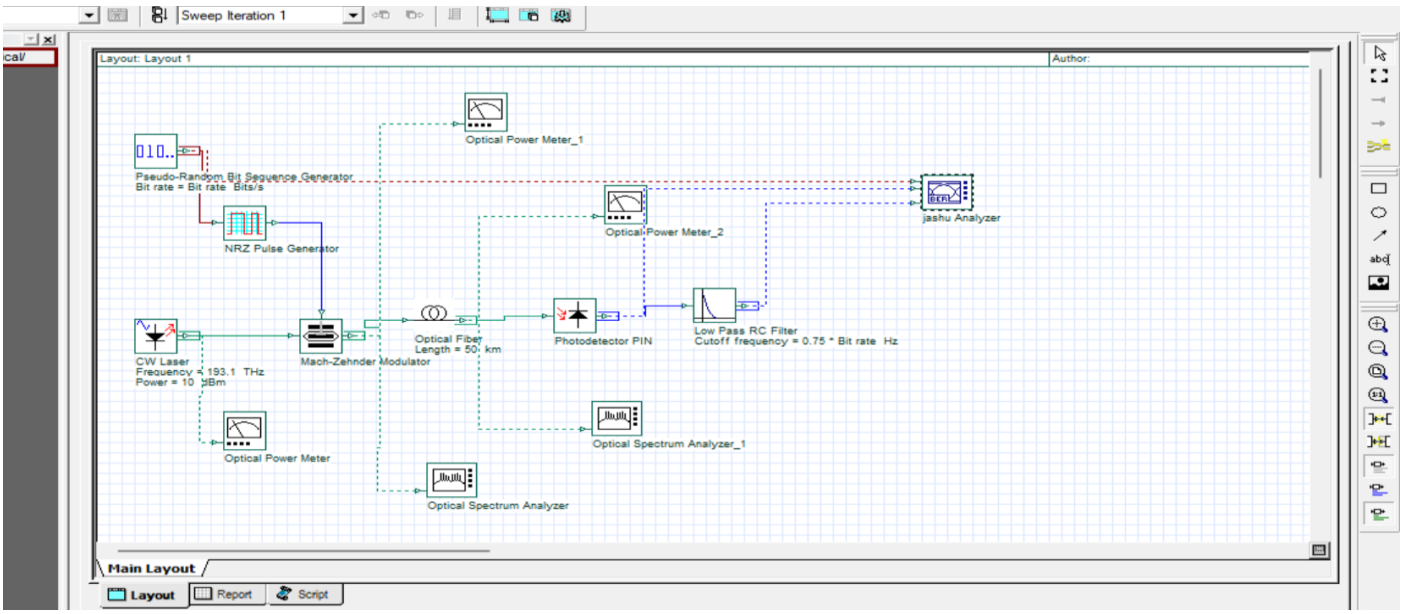
OBJECTIVES

1. Design and establish an optical fiber communication link using OptiSystem software.
2. Calculate and evaluate the Bit Error Rate (BER) for different optical source power levels.
3. Analyze how variation in optical source power affects BER and overall link performance.

SOFTWARE REQUIRED

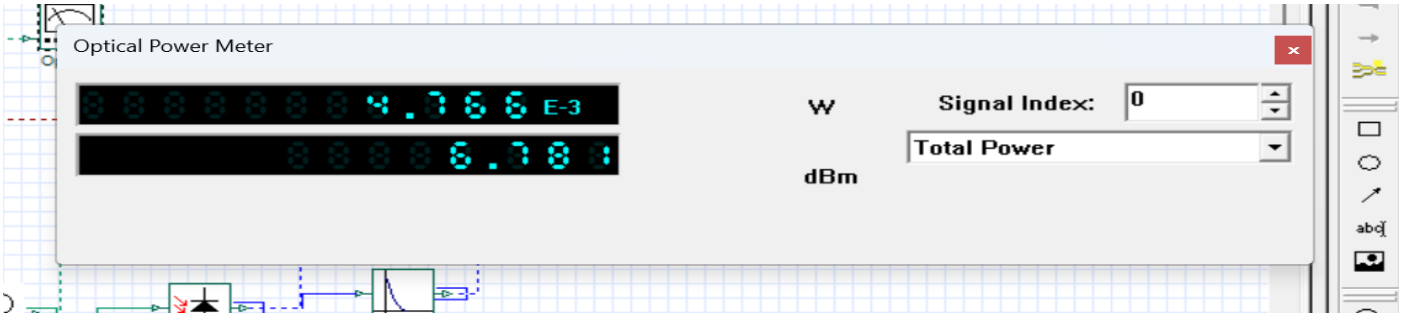
- Optisystem Software

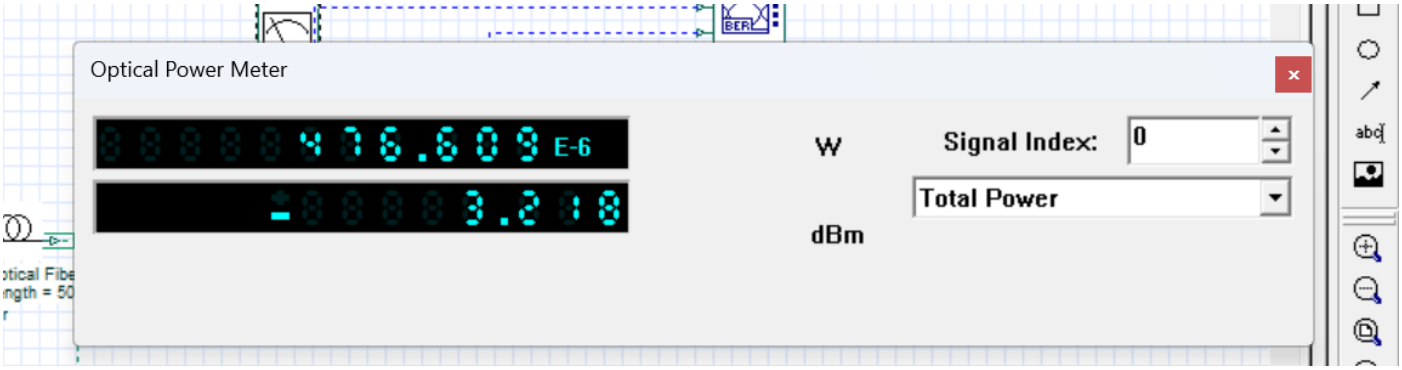
BLOCK DIAGRAM

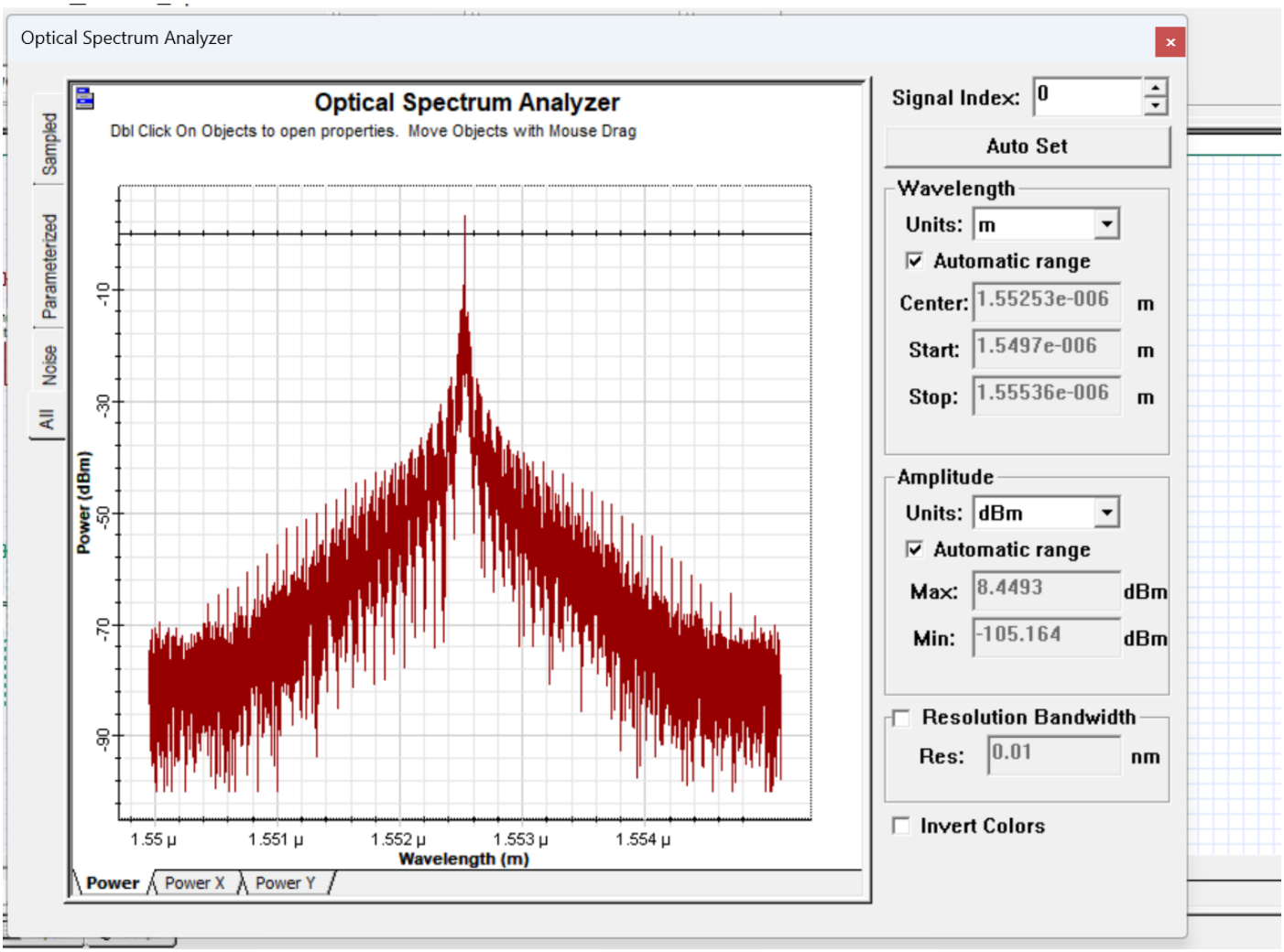


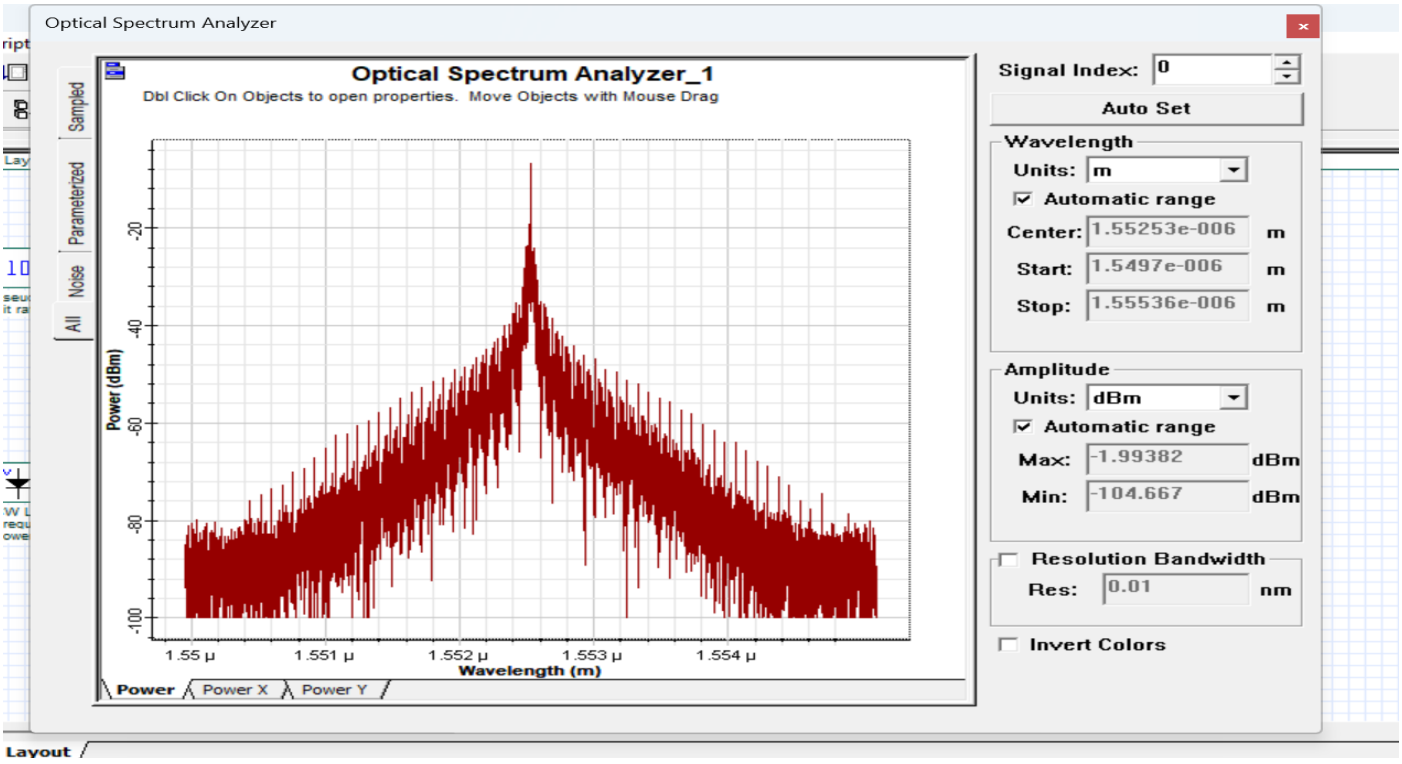
SAMPLE OUTPUT

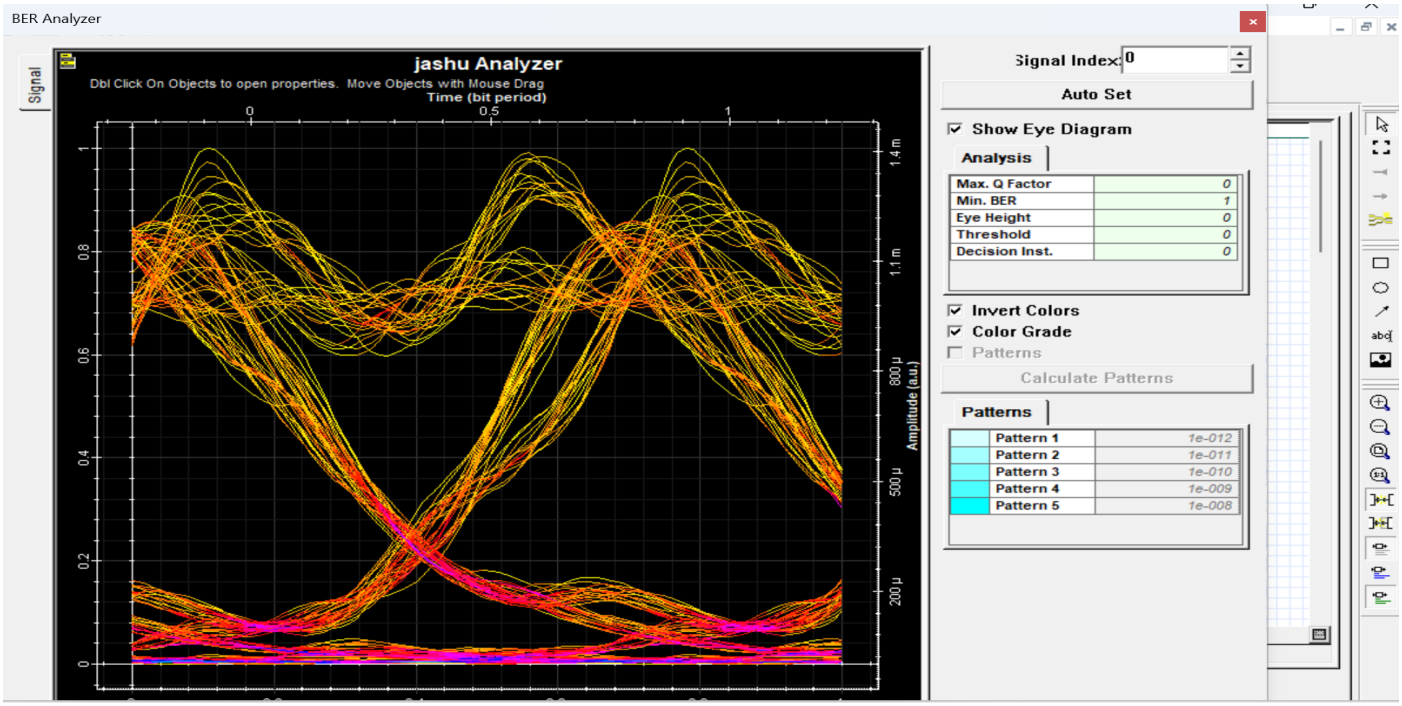



Optical Power Meter
4.766 E-3
6.781
W
dBm
Signal Index:
0
Total Power


Optical Power Meter
476.609 E-6
3.210
W
dBm
Signal Index:
0
Total Power


Optical Spectrum Analyzer
Optical Spectrum Analyzer
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Sampled
Parameterized
Noise
All
Power (dBm)
-10
-30
-50
-70
-90
1.55 µ
1.551 µ
1.552 µ
1.553 µ
1.554 µ
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: 8.4493 dBm
Min: -105.164 dBm
Resolution Bandwidth
Res: 0.01 nm
Invert Colors





Reflection: I learned how to calculate the power requirements of an optical link. I understood how BER helps evaluate the reliability of data transmission.

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LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighta ge
System Design & Link Setup	Correct selection and connection of OptiSystem components for the optical fiber link (PRBS Generator, NRZ Pulse Generator, CW Laser, Mach-Zehnder Modulator, 50 km Optical Fiber, PIN Photodetector, Low Pass RC Filter, BER Analyzer).	All transmitter, modulator, fiber, receiver, and analyzer components are correctly selected, connected in sequence, and configured without errors.	All components are present with minor connection or layout errors, but the link remains functional.	Component selection is partially complete; minor connection or configuration errors impede signal flow.	Incorrect component selection or improper setup leading to complete simulation failure.	20%
Parameter Configuratio n & Execution	Accurate setup of input parameters: CW Laser frequency (193.1 THz), optical fiber length (50 km), and laser power levels (-10 dBm, 0 dBm, 10 dBm).	All input parameters and power-level test cases are accurately set and simulated.	Test case parameters are mostly accurate with minor setting discrepancies.	Basic parameters are configured, but one or more power levels are missing.	Incorrect parameter settings or failure to run the required power cases.	20%
Measuremen t & BER Observation	Accurate extraction and recording of Max. Q-Factor, Min. BER, Eye Height, Threshold, and Decision Limit from the BER Analyzer for each laser power.	Precise data extraction for all power levels, neatly tabulated, with clear eye diagrams and BER plots captured.	Data is collected with minor omissions in Q-factor, BER, or eye parameters.	Partial data collection with incomplete tables or missing eye diagrams.	Inaccurate, incomplete, or missing data and measurements.	20%
BER & Power Level Analysis	Detailed evaluation of the effect of optical source power on BER, Q-factor, and eye opening, including receiver sensitivity and power margin.	Comprehensive analysis correctly relating laser power to received signal strength, Q-factor, BER improvement, and saturation or nonlinear effects.	Clear analysis identifying basic trends in Q-factor and BER versus laser power.	Basic attempt at result analysis without explaining physical mechanisms such as insufficient received power or nonlinearity.	Erroneous or missing analysis of simulation results and eye diagrams.	20%
Report & Conclusion	Identification of the optimum optical source power for reliable communication, comparison of low, moderate, and high power cases, and structural clarity.	Report features clear eye diagrams, complete output tables, and insightful conclusions identifying the optimum laser power.	Satisfactory report with relevant eye diagrams and sound concluding summary.	Incomplete report with missing diagrams, tables, or weak conclusions.	Disorganized or incomplete report lacking required diagrams and valid conclusions.	20%

Result: A fiber-optic link was designed, and link budget and BER were analyzed successfully.